2.3.4 $(a+b)^2 = a^2 + 2ab + b^2$

$(a+b)\cdot(a+b) = a(a+b) + b(a+b) = a\cdot a + a\cdot b + a\cdot b + b\cdot b$

$= a^2 + ab + ab + b^2$

$= a^2 + 2ab + b^2$

Intermediate details:

$n^2 = n^1 \cdot n = n^0 \cdot n \cdot n = n \cdot n$ by definition

$2(nm) = 1(nm) + nm = nm + nm$ by definition